STUDEXPO-2026
Mobile apps and artificial intelligence

AgroZen

An AI mentor in your smartphone for beginner poultry farmers

Working MVP · Android | **3 languages** | **Works offline**

Smart barn sensors | **Reports for banks**



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2017
Low-income families got loans to raise chickens. Reportedly, many borrowers couldn't repay, and banks began cutting such loans.

From 1 August 2026
The state launched concessional poultry loans: up to 10 years at 10% a year.

THE PROBLEM

Loan taken, flock lost

No knowledge
A beginner doesn't know the norms for temperature, feeding and vaccination.

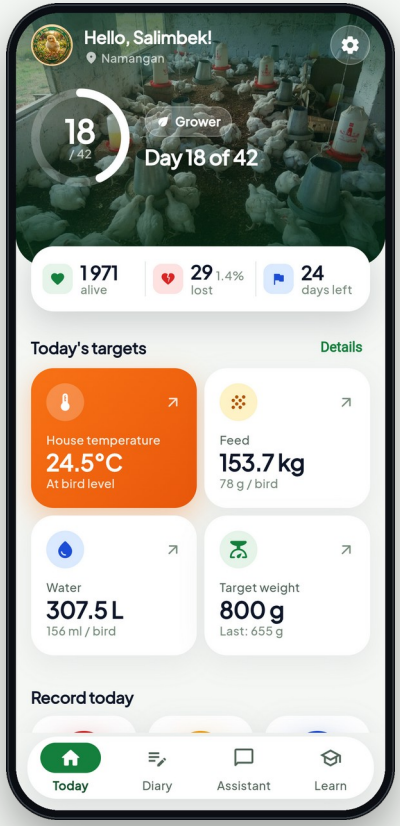
The first 14 days are the most dangerous
A microclimate mistake leads to mass mortality — and nobody measures ammonia or CO2.

The specialist is far away
There is no vet nearby at the moment a question comes up.

The birds are gone. The debt remains.
People lose money not because of laziness, but because of mistakes no one warned them about.

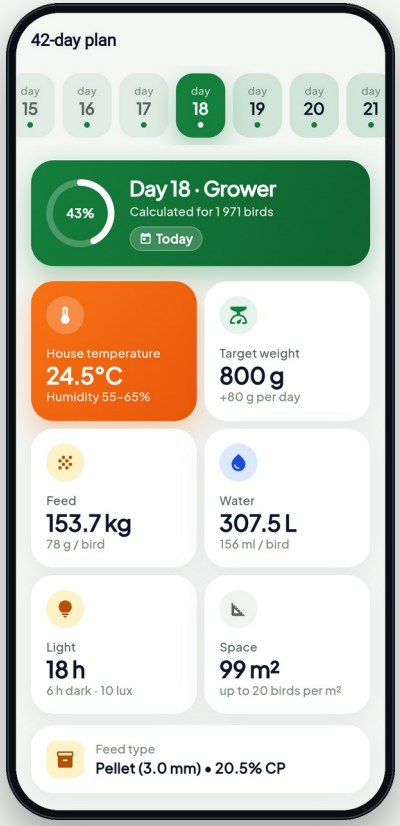
THE SOLUTION

AgroZen guides the farmer through all 42 days



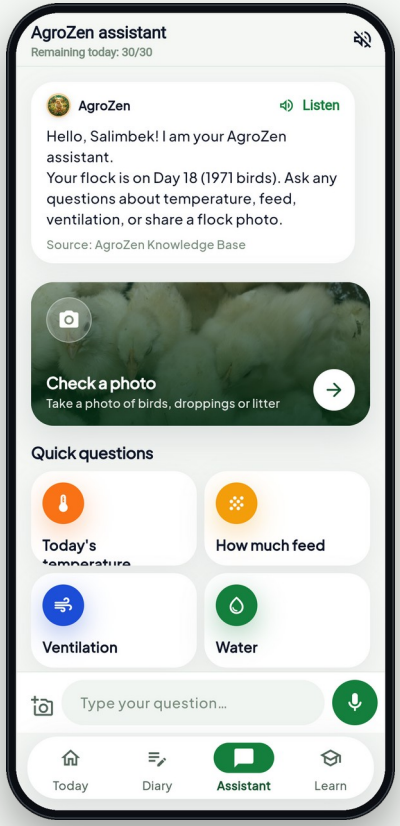
Today

Targets for temperature, feed and water



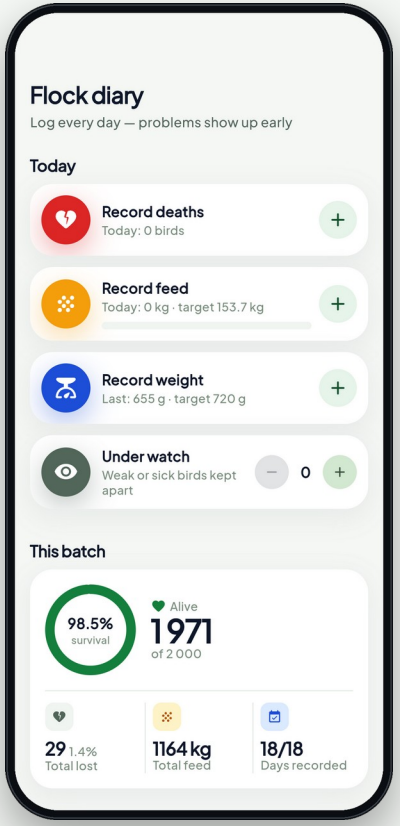
42-day plan

Ross-308 / Cobb-500 norms for every day



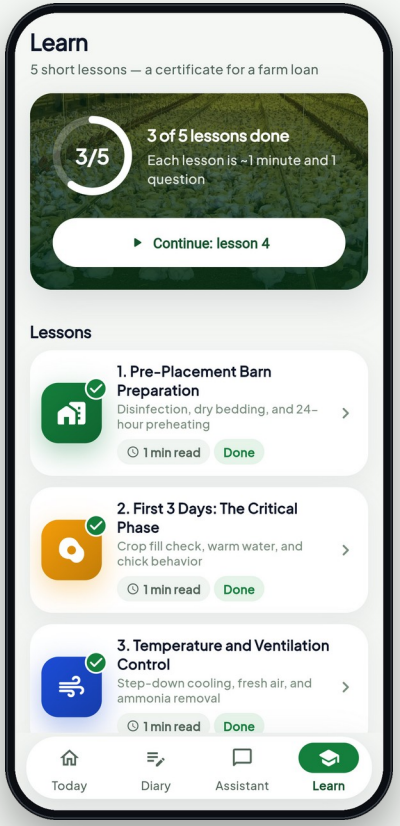
AI assistant

Voice, text or photo — in your language



Flock diary

Losses above the norm — the vet's number

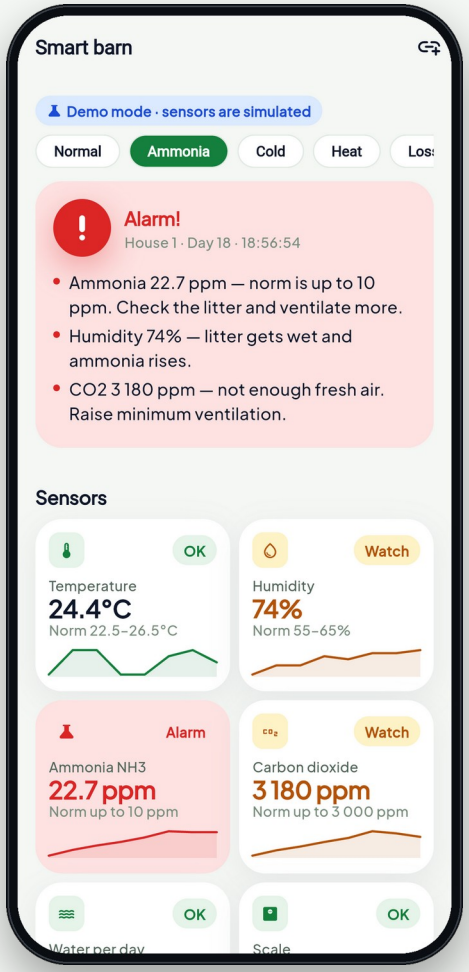


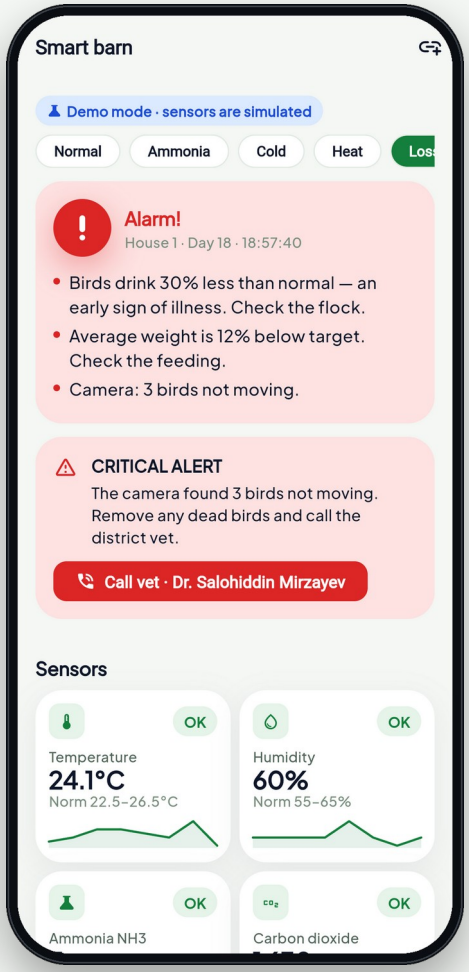
Course

5 lessons and a certificate for a loan

NEW · SMART BARN PROTOTYPE

Sensors and an AI camera watch the flock day and night







AI camera

Birds that stop moving, huddling, heat stress

NH3 up to 10 ppm
People smell it only at 20

CO2 up to 3 000 ppm
Minimum ventilation

Fans switch on
Automatically, by relay

Sensors
ESP32: temperature, humidity, NH3, CO2, water, scale

→

Edge AI
Jetson Orin Nano + IP cameras, video stays on the farm

→

Norms
Ross-308 / Cobb-500 for the day — plain code, not AI

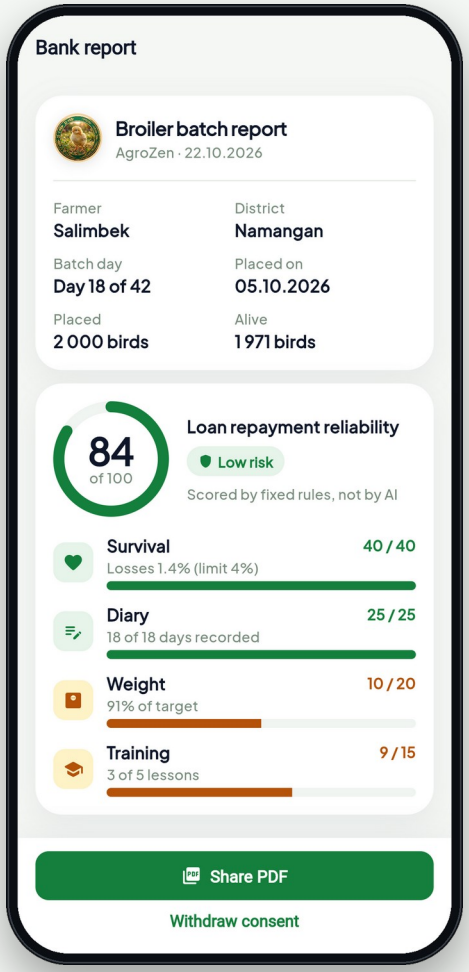
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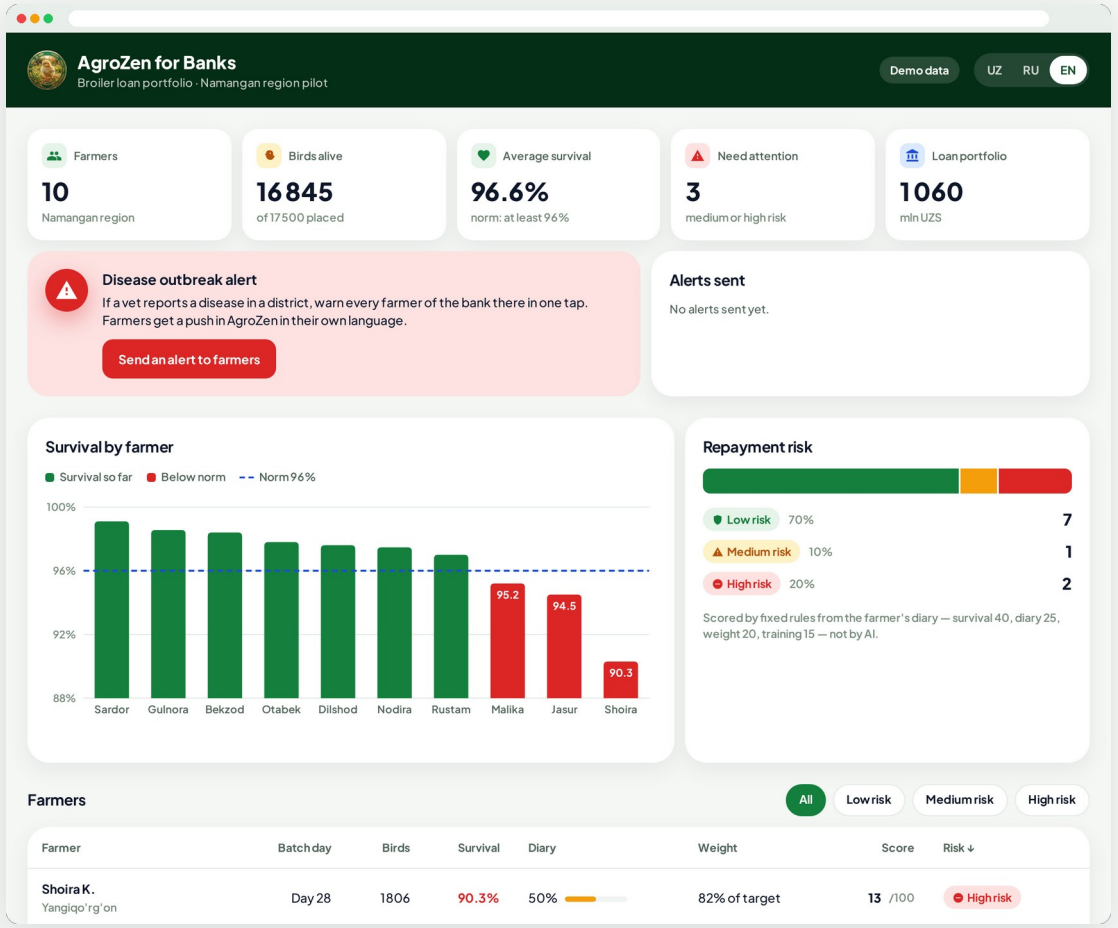
Action
Alert, the vet's number, fans on

Demo mode today: readings are simulated. Hardware goes into two pilot farms in 2027.

NEW · FOR BANKS

Banks see how the flock is doing — and trust the loan





Bank dashboard: 10 pilot farmers, survival against the norm, risk and a district disease alert

Farmer's consent
The report is built from the diary and sent as a PDF only after the farmer agrees.

Score 0–100
Survival, diary, weight, training — fixed rules the bank can check, not AI.

Disease alert
One tap warns every farmer of the bank in a district.

Free for farmers. Banks and microfinance pay: a trained farmer is more likely to repay.

THE RESULT

Not a concept: the app already works

42 days
growing plan with norms for every day

5 lessons
a course with tests and a certificate

13 regions
vet contacts across Uzbekistan

260 tests
automated tests check norms, alerts and scores

The AI answers only from a verified knowledge base



Question
voice, text or photo

Search the base
Ross-308 / Cobb-500

Not in the base?
"ask your vet"

Alerts are counted by plain code against the norms — they never depend on an AI answer.

Flutter
Android, iOS, web

Supabase + Gemini
cloud data and AI

Offline
plan and diary

3 languages
uz · ru · en

No diagnoses or dosages — the app calls the district vet in time.

NEXT STEP

A pilot with 10 farmers and a bank in Namangan region

92–95%
flock survival

80%+
days with diary entries

96%+
AI answers matching the norms

Target figures per flock (42 days).

- Sensors and AI cameras in two pilot farms
- Bank dashboard with real pilot data
- Then laying hens, sheep and cattle, greenhouses



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Download and try it
Android app
github.com/ZedRubilan/agrozen

Thank you!